Enhancing Incident Response Through Sigma MCP and Operational Policy Integration with LLM

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***Abstract*—In this study, we propose Sigma MCP (Model Context Protocol), a method designed to improve the accuracy of incident response in computer security operations. The proposed system detects suspicious logs based on Sigma rules and analyzes the results using a large language model (LLM) to enable high-precision attack assessment. While logs that match Sigma rules are generally considered highly indicative of attacks, our approach extends this by incorporating organization-specific operational rules and security policies, which are interpreted and evaluated by the LLM to further improve accuracy.**

In recent years, attack techniques such as Living off the Land (LotL), which abuse legitimate tools and commands, have become increasingly prevalent. This trend makes it more difficult to distinguish malicious activity from normal operations. In such environments, reducing false detection has become a critical challenge. To address this issue, Sigma MCP combines rule-based deterministic detection with the flexible interpretation of operational knowledge via an LLM. This design enables even non-expert security personnel to effectively reduce the overhead of false positive handling.

To evaluate the effectiveness of the proposed approach, we conducted a comparative analysis with Elasticsearch MCP. The results showed that when detection was performed using Sigma MCP with operational rule consideration, the identified events were more likely to be actual attacks, and the overall detection precision improved. In contrast, while Elasticsearch MCP was able to detect a wider range of events, it was also more prone to generating false positives.

***Index Terms*—Sigma Rules, Large Language Model (LLM), False Positive Reduction, Model Context Protocol (MCP), Incident Response**

I. Introduction

In the field of cybersecurity, a severe shortage of skilled personnel and a mismatch between required and available expertise—commonly referred to as the skill gap—have become major challenges [1]. Even under such constraints, when a cybersecurity incident occurs in another organization, the relevant information is often shared with others. Organizations receiving such information are required to quickly assess whether similar attacks have occurred within their own environments and implement countermeasures if necessary, even if they lack sufficient technical knowledge or experience [2].

Moreover, as seen in techniques like Living off the Land (LotL), the increasing prevalence of attacks that abuse legitimate tools and operations has made it increasingly difficult to distinguish malicious activities from normal operations [3]. These circumstances place a significant burden on organizations with limited security response capabilities. As such, the skill gap becomes particularly apparent during incident response, where technical knowledge and situational judgment are essential.

To support organizations affected by the skill gap in conducting timely and accurate incident response, this study proposes Sigma MCP (Model Context Protocol), a support framework that integrates Sigma rules with large language models (LLMs). Sigma rules provide a mechanism for detecting malicious activity in logs based on information described in threat intelligence reports, such as tools or techniques used by attackers. They offer a standardized detection-as-code format, enabling security teams to efficiently define and share detection logic across different platforms [4]. Importantly, Sigma has been recognized as a generic and open signature format for SIEM systems and can be automatically converted into queries for widely used SIEM products such as Splunk or Elasticsearch. These rules have already been adopted by many SIEM (Security Information and Event Management) platforms [5]. In many cases, threat reports include exploited tools and TTPs (Tactics, Techniques, and Procedures), allowing for detection of similar attacks using corresponding Sigma rules [6].

The proposed Sigma MCP framework enhances traditional Sigma-based detection by incorporating an AI agent powered by an LLM. This design enables users to interact with the system in natural language during investigation, facilitating flexible incident handling regardless of the user's technical expertise. Additionally, Sigma MCP aims to reduce false positives by allowing the LLM to interpret organization-specific security policies and operational rules.

To evaluate the effectiveness of this approach, we designed attack scenarios that could be detected using Sigma rules and executed mock attacks in a test environment that assumed

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a specific condition. The target environment was based on Windows and Active Directory, which are widely adopted across organizations. This configuration was selected due to its generality and practicality; Active Directory is reportedly used by approximately 90% of Fortune 1000 companies [7].

In the evaluation, we verified whether Sigma MCP could accurately detect the attacks and whether the LLM could generate appropriate responses while considering the organization's operational rules. For comparison, we also evaluated Elasticsearch MCP to assess its detection performance against the same set of attacks.

II. Related Technologies

A. Log Analysis Using Sigma Rules

Sigma is a generalized, structured rule format designed for writing detection rules against log data. Sigma rules are written in a highly readable YAML format and are designed to be platform-independent, allowing them to be applied across various SIEM and log management systems. Each rule typically specifies the conditions of the target logs (such as log source, event ID, and field values), along with descriptive metadata such as the rule's purpose, associated tags, and detection logic. These rules can be converted into the query languages of various SIEM platforms (e.g., Elasticsearch DSL, Splunk SPL) through supporting tools.

The Sigma project is maintained and continuously expanded by the security community, with a large number of publicly available rules mapped to tactics and techniques from the MITRE ATT&CK framework. When a log entry matches a Sigma rule, it is likely to indicate malicious activity. Because each rule also includes a human-readable description, it is easier for analysts to understand the nature of the detected attack. Additionally, many Sigma rules describe conditions under which false positives may occur, enabling LLMs to take such context into account during analysis.

However, applying all available Sigma rules at once is generally not recommended due to resource constraints [8]. Therefore, it is important to selectively identify and apply only the relevant rules during log analysis.

B. Model Context Protocol (MCP)

The Model Context Protocol (MCP) is an open standard that defines a bidirectional interface between large language models (LLMs) and external tools or data sources [9]. Inspired by the Language Server Protocol, MCP introduces a client–server architecture in which hosts can dynamically discover and orchestrate tools provided by servers. Recent studies emphasize that MCP reduces fragmentation in agent architectures and establishes a secure foundation for interoperable, AI-driven workflows [10].

III. Related Work

A. Utilization of Sigma Rules

As a study on the integration of Sigma rules with AI, a method has been proposed in which log events detected by Sigma rules are passed to Splunk UBA, allowing AI to analyze the context and correlation and score anomalies. In addition, by combining with Elastic EQL, it becomes possible to analyze causal relationships among multiple logs and detect attack sequences, which are difficult to achieve with Sigma rules alone. These applications demonstrate the advanced use of Sigma rules. These studies indicate that Sigma rules, when integrated with AI and existing log analysis platforms, can enable more flexible and accurate threat detection [11].

B. Suppression of False Positives

To suppress false positives, some approaches have AI learn from analysts' past decisions and feedback to automatically evaluate the content and context of alerts, thereby reducing false positives. In particular, the use of generative AI has been proposed to identify erroneous alerts and prioritize responses based on severity [12]. Furthermore, some articles suggest the possibility that command input for system maintenance, such as on a Windows system, could be mistakenly detected as an attack, i.e., a false positive [13]. False positives in Security Operation Centers (SOCs) require time to investigate nonexistent threats [14]. Therefore, in our proposed method, we examine whether sharing operational rules with LLMs each time could enable analysis that takes such rules into account.

IV. Proposed Method

This study proposes a system that determines the presence of attack traces by cross-referencing user-provided detection requests, written in natural language, with log data that match corresponding Sigma rules. The system further validates consistency with organizational operational policies (i.e., organization-specific rules that define what activities are considered acceptable or not). To achieve this, we build a system centered around the Sigma MCP Server.

To evaluate the behavior of the Sigma MCP, we implement the system with the following components:

- MCP Client: GitHub Copilot in Visual Studio Code
- LLM1 (Language model used by the MCP Client): Claude Sonnet 4
- MCP library for Sigma: fast MCP [15]
- LLM2 (Language model used by the Sigma MCP Server): gpt-4.1-2025-04-14
- Log Analytics Platform: Elasticsearch 7.7.0

For the experiments using gpt-oss, the model was executed locally on an Apple M3 Pro chip with 36 GB unified memory, ensuring that no queries were transmitted to external cloud services.

A. Behavior of Sigma MCP

When a request is made by the user, the MCP Client may connect to the Sigma MCP Server. As illustrated in Figure1, the workflow is as follows: a user submits a natural language request, which is processed by LLM1 in the MCP Client. If necessary, LLM1 forwards the request to the Sigma MCP Server, where LLM2 retrieves relevant Sigma rules and converts them into executable queries for the log analytics platform. Operational policies are then incorporated into the results, and LLM1 provides the final response to the user.

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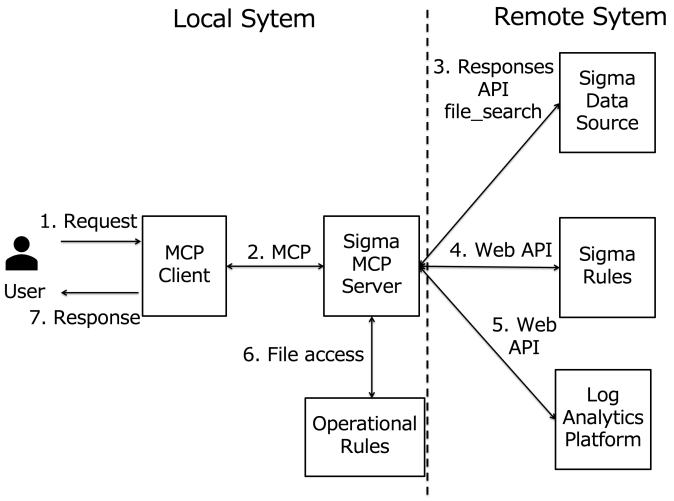


Fig. 1. Overview of the proposed method

B. Supplementary Details on the Process

During implementation, we observed that not all Sigma rule references retrieved using the `web_search` function (a feature of the OpenAI Responses API [16]) were valid or accessible. Out of 100 retrieved URLs, only 24 were valid and usable, with the rest resulting in HTTP 404 errors. This limitation highlights the practical challenge of relying on community-maintained repositories. To address this issue, we created and uploaded a pre-processed JSON index containing the filename, description, and URL of Sigma rules, and used the `file_search` function [16] to ensure that only valid URLs were retrieved.

C. Example of Sigma MCP Behavior

A concrete investigation scenario is presented below. In this case, the user conducting the investigation is assumed to have limited technical expertise. The situation involves the user reading a security report, extracting keywords of interest, and checking whether those keywords indicate potential malicious activity in their organization. As a specific example, we illustrate a case in which the user receives information that the tool secretsdump may be exploited for credential theft.

- The user submits a request to the MCP Client: ”I want to detect secretsdump abuse using Sigma.”
- Claude Sonnet 4, integrated with GitHub Copilot, determines that it needs to connect to the Sigma MCP Server and converts the request into a suitable input format. An example input is: ”query_text”: ”impacket secretsdump.py”, ”rule_count”: 5. Here, rule_count specifies the number of Sigma rules to be searched.
- Based on the input, LLM2 selects appropriate Sigma rules and retrieves their URLs [17] [18] [19] [20].
- The Sigma MCP Server accesses the obtained URLs and downloads the corresponding Sigma rule YAML files.
- These YAML rules are converted into es-dsl, a JSON-based query language used to interact with Elasticsearch. If matching logs are found, the search results are returned; if not, the response is ”No results found for the given Sigma rule.”
- Along with the search results, operational policies, as illustrated in Figure 2 are appended.
- LLM1, integrated with the MCP Client, analyzes both the operational policies and the search results to generate a response to the user’s request.

```
System Role Mapping
* Domain Controller (dc2019.example.local): 192.168.3.30
* Maintainer PC (maintainer-pc.example.local): 192.168.3.31
* File Server (file2019.example.local): 192.168.3.32

Operational Rules
* Regular working hours are from 9:00 AM to 6:00 PM (JST).
* Only the Maintainer PC is allowed to execute administrative commands on the Domain Controller and File Server via RDP or PsExec.
* The Maintainer PC must be accessed using the domain administrator account named dcadmin. No other accounts are used.
* Domain Controller and File Server do not access the internet
```

Fig. 2. Example of operational policies

V. Comparative Evaluation of Sigma MCP and Elasticsearch MCP

A. Evaluation Environment

To conduct a comparative evaluation between Sigma MCP and Elasticsearch MCP, we built a simplified Active Directory test environment as shown in Figure 3. Client PCs 01, 02, and 03, along with the Maintenance PC, belong to the same domain named example.local. Only the Maintenance PC is allowed to execute administrative commands on the Domain Controller and access the Entra ID management console for operational purposes. These conditions are defined as operational policies, as illustrated in Figure 2, and are provided to the Sigma MCP Server.

In this environment, both the Domain Controller and the Maintenance PC are configured to generate specific security-related event logs, as summarized in TableI. The event logs are forwarded to the Elasticsearch platform via Winlogbeat for log collection and analysis.

We collected Windows event logs, including both native Security logs and Sysmon logs, for approximately 17 days and transferred them using Winlogbeat from both the Domain Controller and the Maintenance PC. A total of 244,676 events were obtained, of which 81.1% originated from the Domain Controller and 18.9% from the Maintenance PC. The

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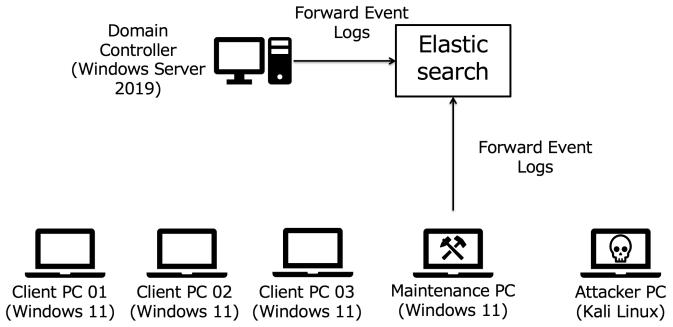


Fig. 3. Overview of the evaluation environment

TABLE I
LIST OF COLLECTED EVENT IDS

Event ID	Summary	Remark
1	Process creation	Sysmon is requried
4688	A new process has been created	Command line process auditing is requried
5145	A network share object was checked	-
11	FileCreate	Sysmon is requried

logs included event types listed in TableI, such as process creation and file creation, as well as authentication requests, logon/logoff events, and access to shared resources.

B. Comparative Evaluation Methodology

In these comparative experiments, we varied LLM1 (the language model used by the MCP Client) while keeping LLM2 (the language model used by the Sigma MCP Server) fixed. Specifically, we evaluated multiple LLMs as LLM1, including Claude Sonnet 4, GPT-4o, and gpt-oss-20b [21]. For each setting, LLM2 remained constant. Claude Sonnet 4 was also evaluated independently without MCP integration. The inclusion of gpt-oss is particularly relevant because it operates as a local LLM and does not transmit queries to external cloud services. This makes it more suitable for scenarios where sensitive or confidential information must be processed securely within an organization, and it also has the advantage of incurring no per-request cost.

To evaluate the behavior of Sigma MCP, we conducted a series of tests using widely known offensive tools against test Windows machines and observed whether detection was successful. As a comparison, we also included evaluations of Elasticsearch MCP [22], which is publicly provided by Elasticsearch Inc., under the same LLM1 settings. The following prompts were issued in each setting:

- For Sigma MCP with Claude 4, we queried the MCP Client with the string: “We want to detect the abuse of [TOOL] using Sigma.”
- For Elasticsearch MCP with Claude 4, we used: “We want to detect the abuse of [TOOL] using Elasticsearch.”
- For Claude Sonnet 4 alone, we queried the Anthropic API with: “Please create an es-dsl query to detect the abuse of [TOOL].”

Here, [TOOL] was systematically replaced with the name of each offensive tool under evaluation, namely: AADInternals, BloodHound, Mimikatz, Rubeus, secretsdump.py, psexec.py, smbexec.py, and wmiexec.py.

The selection of these tools was based on a recent analytical report [23] that discusses detection techniques in Windows, Active Directory, and Entra ID environments. Only tools that are freely available without requiring user registration were chosen, specifically: AADInternals, BloodHound, Mimikatz, Rubeus, and four tools from the Impacket suite—secretsdump.py, psexec.py, smbexec.py, and wmiexec.py.

Whenever possible, attack commands were deliberately crafted to generate logs that would match existing Sigma rules. This approach was taken to evaluate the effectiveness and response precision of Sigma MCP based on known log patterns, rather than to assess the accuracy of the Sigma rules themselves.

C. Evaluation Results with Claude Sonnet 4

Table II presents a comparison of the detection results among Sigma MCP, Elasticsearch MCP, and Claude Sonnet 4 used independently. The numbers listed in the ”Sigma MCP with Claude4” column indicate the matched Sigma rules. Focusing first on Sigma MCP and Elasticsearch MCP, both methods successfully detected tools such as AADInternals, Mimikatz, and Rubeus by identifying characteristic keywords present in the command line field of Windows Event ID 4688. Here, malicious tools with the .exe extension (sharphounds.exe, mimikatz.exe, rubeus.exe) were renamed to svchost.exe before execution.

TableII also allows us to derive quantitative metrics for detection performance. For Sigma MCP (TP = 7, FN = 1, FP = 0), the values are Precision = 1.0, Recall = 0.875, and F1-score = 0.93. In comparison, Elasticsearch MCP with Claude4 (TP = 4, FN = 2, FP = 2) achieved Precision = 0.67, Recall = 0.67, and F1-score = 0.67. These results quantitatively confirm that Sigma MCP provides higher precision with fewer missed detections, whereas Elasticsearch MCP shows a less balanced trade-off between precision and recall.

A key difference was observed in their handling of false positives. Elasticsearch MCP employed a broader search strategy, such as matching general keywords like “powershell” and then checking for related activity. While this approach increased detection breadth, it also led to false positives in legitimate use cases. For instance, when PsExec was used for normal administrative tasks, Elasticsearch MCP flagged the appearance of PSEXESVC.exe as suspicious, despite being part of a sanctioned operation. The specific causes of false positives are noted in parentheses within the table.

In contrast, Sigma MCP relies on precise Sigma rule patterns that describe specific attack behaviors, such as Impacket’s psexec.py or wmiexec.py. This enabled it to distinguish between malicious and benign uses of similar tools. Furthermore, Sigma MCP incorporates operational policies into the detection process. When PsExec was executed, the

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TABLE II
EVALUATION RESULTS

Tool name	Sigma MCP with Claude4	Elasticsearch MCP with Claude4	Claude Sonnet 4	Attack Command
AADInternals	True Positive [24]	True Positive	False Negative	powershell.exe -ExecutionPolicy Bypass -Command "Get-AADIntSyncCredentials -?"
BloodHound	True Positive [25]	True Positive	Invalid query	svchost.exe exection
Mimikatz	True Positive [26]	True Positive	Invalid query	.\svchost.exe privilege::debug sekurlsa::logonpasswords (Renamed mimikatz.exe to svchost.exe)
Rubeus	True Positive [27]	True Positive	Invalid query	.\svchost.exe ptt /ticket:<base64-encoded ticket> (Renamed mimikatz.exe to svchost.exe)
impacket(secretdump)	True Positive [17]	False Negative	False Positive (legitimate log-off, Event ID 4634)	secretsdump.py '<domain name>/<ID>: <Password>'@<Target IP address>
impacket(psexec)	True Positive [28]	False Positive (Legitimate PsExec.exe usage, Event ID 4688)	False Positive (Legitimate PsExec.exe usage, Event ID 5145)	psexec.py '<domain name>/<ID>:<Password>' @<Target IP address>whoami
impacket(smbexec)	True Positive [29]	False Positive (Legitimate schtasks.exe usage, Event ID 4688)	False Positive (Legitimate log-on, Event ID 4624)	smbexec.py '<domain name>/<ID>:<Password>' @<Target IP address>
impacket(wmiexec)	False Negative	False Negative	Invalid query	wmiexec.py '<domain name>/<ID>:<Password>' @<Target IP address>

system checked whether the initiating host was an authorized maintenance PC. If not, the LLM-generated output noted the inconsistency and recommended further verification.

For tools like Mimikatz and Rubeus, which are rarely used in legitimate environments, Sigma MCP raised alerts regardless of the originating host, demonstrating its ability to incorporate domain knowledge in context-aware assessments.

For Claude Sonnet 4 without MCP, we also constructed queries using characteristic keywords of offensive tools. However, many of these were rejected as invalid queries. In contrast, Elasticsearch MCP exhibited behavior that automatically resolved such errors, for example, by inspecting the index or correcting mistakes in the query. It is possible that, if we had instructed the LLM to correct the invalid queries, it might have provided properly functioning ones. However, we did not give such instructions or make any manual corrections. Even when the queries were not marked as invalid, Claude Sonnet 4 often detected completely unrelated activities, such as Log-off or Log-on events, or raised false positives by flagging operational use of PsExec. As a result, no true positives were observed.

As a result, Sigma MCP successfully suppressed false positives. This demonstrates the effectiveness of combining rule-based detection with policy-aware LLM reasoning.

D. Evaluation Results with Other Models

When GPT-4o was used as LLM1, the Sigma MCP system often failed to locate appropriate Sigma rules. The agent tended to search only with the exact keywords provided by the user. For example, given the query "We want to detect the abuse of mimikatz using Sigma," the agent issued the query: `"query_text": "mimikatz"` This simple query failed to retrieve relevant rules. In contrast, Claude Sonnet 4 occasionally retried when the first attempt failed and automatically expanded the query. For instance, in the second attempt, it used: `"query_text": "mimikatz credential dumping sekurlsa logonpasswords dcsync LSASS memory dump"` This behavior enabled Claude Sonnet 4 to find the correct rules more consistently. As a result, GPT-4o produced only one true positive (BloodHound abuse) and failed to detect the other cases (all false negatives).

In the case of gpt-oss, the system did not trigger Sigma MCP calls in any of the tested scenarios. Consequently, all attempts resulted in false negatives. While gpt-oss has the advantage of being deployed locally and not sending queries to external services, our evaluation shows that its current performance in Sigma rule retrieval is limited. Since no Sigma MCP calls were issued, we did not conduct additional experiments using gpt-oss-20b as LLM2.

E. Limitations and Future Work

This study has several limitations. First, the evaluation was conducted using attack commands that were intentionally crafted to produce logs matching existing Sigma rules. While this design helps validate the feasibility of Sigma MCP, it does not fully test the robustness of the system against adversaries who deliberately attempt to evade detection. Second, Sigma MCP fundamentally depends on predefined Sigma rules. Consequently, it cannot detect novel or zero-day techniques that are not covered by existing rule sets. Third, the evaluation environment was intentionally simplified to clearly observe the behavior of the system, but this controlled setup lacks the noise, diversity, and complexity of real-world enterprise networks. These limitations highlight that the current study focuses on demonstrating the concept and feasibility of Sigma MCP rather than proving its effectiveness across all operational scenarios.

Future research will extend the evaluation of Sigma MCP to larger-scale and more diverse environments that better reflect real-world conditions. In particular, experiments incorporating realistic user activities, heterogeneous system configurations, and background noise will help to validate the robustness and scalability of the proposed approach.

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VI. Conclusion

From the results, it was observed that using Elasticsearch MCP enables broader detection coverage compared to the proposed method. While false positives may occur, we believe that adherence to operational policies and the practice of recording and cross-referencing daily activities make it possible to appropriately determine whether an alert is a false positive.

On the other hand, the proposed method is based on Sigma rules and, by design, cannot detect events that do not match any predefined rules. However, if a log is detected, the likelihood of it being a false positive is low. This is because many Sigma rules explicitly describe conditions under which false positives may occur, and the proposed method incorporates LLMs to interpret operational policies, allowing potential false positives to be taken into account during analysis.

Therefore, the proposed method should not be used as a standalone solution for attack detection, but rather as a supplementary tool to support more comprehensive detection strategies.

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